

ECO 403/503 Midterm Project – Solution Notes

How to Lose Your Degree Because of Cheating: A Game-Theoretic Analysis of NU's Academic Disciplinary Committee

Throughout, let:

- $\theta \in \{G, I\}$ be the true state (Guilty / Innocent),
- $s_i \in \{A, E\}$ be juror i 's private signal (Accusatory / Exculpatory),
- $a_i \in \{0, 1\}$ be juror i 's action (0 = acquit, 1 = convict),
- $\mu = \Pr(\theta = G) \in (0, 1)$ be the common prior on guilt,
- $p \in (1/2, 1)$ be the probability of an accusatory signal if $\theta = G$,
- $q \in (1/2, 1)$ be the probability of an exculpatory signal if $\theta = I$,
- ξ be the correlation parameter in the signal distribution.

The committee convicts if and only if both jurors vote to convict, i.e. $(a_1, a_2) = (1, 1)$.

Jurors' payoffs:

- Correct verdict (guilty convicted or innocent acquitted): payoff 0.
- Type I error (innocent convicted): payoff $-z$.
- Type II error (guilty acquitted): payoff $-(1 - z)$.

1.1 Belief Space and Signal Functions

Main solution

A state of the world is $\omega = (\theta, s_1, s_2)$ where

$$\theta \in \{G, I\}, \quad s_1, s_2 \in \{A, E\}.$$

Thus the state space is

$$\Omega = \{G, I\} \times \{A, E\} \times \{A, E\},$$

with $|\Omega| = 8$. Explicitly:

$$\Omega = \{(G, A, A), (G, A, E), (G, E, A), (G, E, E), (I, A, A), (I, A, E), (I, E, A), (I, E, E)\}.$$

The conditional distribution of signals is given by the tables:

For $\theta = G$:

	$s_2 = A$	$s_2 = E$
$s_1 = A$	$p^2 + \xi$	$p(1 - p) - \xi$
$s_1 = E$	$p(1 - p) - \xi$	$(1 - p)^2 + \xi$

For $\theta = I$:

	$s_2 = A$	$s_2 = E$
$s_1 = A$	$(1 - q)^2 + \xi$	$(1 - q)q - \xi$
$s_1 = E$	$(1 - q)q - \xi$	$q^2 + \xi$

The common prior on θ is

$$\Pr(\theta = G) = \mu, \quad \Pr(\theta = I) = 1 - \mu.$$

By the Harsanyi construction, the prior on Ω is

$$\mu(\theta, s_1, s_2) = \Pr(\theta) \Pr(s_1, s_2 \mid \theta).$$

Explicitly:

For $\theta = G$:

$$\begin{aligned}\mu(G, A, A) &= \mu(p^2 + \xi), \\ \mu(G, A, E) &= \mu(p(1-p) - \xi), \\ \mu(G, E, A) &= \mu(p(1-p) - \xi), \\ \mu(G, E, E) &= \mu((1-p)^2 + \xi).\end{aligned}$$

For $\theta = I$:

$$\begin{aligned}\mu(I, A, A) &= (1-\mu)((1-q)^2 + \xi), \\ \mu(I, A, E) &= (1-\mu)((1-q)q - \xi), \\ \mu(I, E, A) &= (1-\mu)((1-q)q - \xi), \\ \mu(I, E, E) &= (1-\mu)(q^2 + \xi).\end{aligned}$$

Each juror i observes only their own signal s_i . The signal (type) functions are therefore:

$$\tau_1(\theta, s_1, s_2) = s_1, \quad \tau_2(\theta, s_1, s_2) = s_2,$$

that is, $\tau_i : \Omega \rightarrow \{A, E\}$ with $\tau_i(\omega) = s_i$.

Diagnostic commentary

LLM Pitfall Alert. A typical mistake is to:

- Only specify $\Pr(\theta)$ and call that the “belief space”, never extending the prior to Ω .
- Or to write the conditional tables but omit the prior $\mu(\theta, s_1, s_2)$ on the full state space.

But the belief space in the Harsanyi sense is (Ω, μ) with μ defined on *all* (θ, s_1, s_2) .

Student-Level Pitfall. Students often describe the signals informally but do not actually define the signal functions τ_i . Here it is important to write $\tau_i : \Omega \rightarrow \{A, E\}$ explicitly.

1.2 Payoff Functions and Strategy Spaces

Main solution

Define the verdict function

$$v(a_1, a_2) = \begin{cases} 1 & \text{if } a_1 = a_2 = 1 \quad (\text{convict}), \\ 0 & \text{otherwise} \quad (\text{acquit}). \end{cases}$$

Juror i 's payoff depends on the true state θ and the verdict $v(a_1, a_2)$:

$$u_i(\theta, s_1, s_2, a_1, a_2) = \begin{cases} 0 & \text{if } \theta = G \text{ and } v(a_1, a_2) = 1, \\ 0 & \text{if } \theta = I \text{ and } v(a_1, a_2) = 0, \\ -z & \text{if } \theta = I \text{ and } v(a_1, a_2) = 1, \\ -(1-z) & \text{if } \theta = G \text{ and } v(a_1, a_2) = 0. \end{cases}$$

Equivalently,

$$u_i(\theta, s_1, s_2, a_1, a_2) = \begin{cases} 0 & \text{if } (\theta = G, a_1 = a_2 = 1) \text{ or } (\theta = I, (a_1, a_2) \neq (1, 1)), \\ -z & \text{if } \theta = I, a_1 = a_2 = 1, \\ -(1-z) & \text{if } \theta = G, (a_1, a_2) \neq (1, 1). \end{cases}$$

A pure strategy for juror i is a function from signals to actions,

$$\sigma_i : \{A, E\} \rightarrow \{0, 1\}.$$

We can represent a strategy as the pair $(\sigma_i(A), \sigma_i(E))$, so each juror has four pure strategies:

$$(0, 0), (0, 1), (1, 0), (1, 1),$$

corresponding to:

- Always acquit,
- Convict only on E (anti-monotone),
- Convict only on A (monotone),
- Always convict.

Diagnostic commentary

LLM Pitfall Alert. Common failures:

- Making payoffs depend directly on the signals (s_1, s_2) instead of on the true state θ and the final verdict.
- Treating a “strategy” as a single action a_i rather than a mapping $\sigma_i : \{A, E\} \rightarrow \{0, 1\}$.

In Bayesian games, strategies are always mappings from types/signals to actions.

Student-Level Pitfall. A frequent error is to forget the unanimity rule and assign payoffs to individual votes rather than to the outcome implied by the pair (a_1, a_2) .

1.3 Expected Payoffs as a Function of Posterior r

Main solution

Let r denote a juror’s posterior probability that the defendant is guilty, given her information:

$$r = \Pr(\theta = G \mid \text{juror's information}), \quad 1 - r = \Pr(\theta = I \mid \text{information}).$$

Expected payoff under acquittal. If the verdict is acquittal ($v = 0$):

- With probability r : $\theta = G$ and the defendant is guilty but acquitted $\Rightarrow$ type II error $\Rightarrow$ payoff $-(1 - z)$.
- With probability $1 - r$: $\theta = I$ and the defendant is innocent and acquitted $\Rightarrow$ correct verdict $\Rightarrow$ payoff 0.

Hence

$$\mathbb{E}[u \mid \text{acquittal}] = r \cdot (-(1 - z)) + (1 - r) \cdot 0 = -r(1 - z).$$

Expected payoff under conviction. If the verdict is conviction ($v = 1$):

- With probability r : $\theta = G$ (guilty, convicted) $\Rightarrow$ payoff 0.
- With probability $1 - r$: $\theta = I$ (innocent, convicted) $\Rightarrow$ type I error $\Rightarrow$ payoff $-z$.

So

$$\mathbb{E}[u \mid \text{conviction}] = r \cdot 0 + (1 - r) \cdot (-z) = -(1 - r)z.$$

Comparison and threshold rule. The difference is

$$\mathbb{E}[u \mid \text{conviction}] - \mathbb{E}[u \mid \text{acquittal}] = -(1 - r)z + r(1 - z) = -z + r.$$

Thus:

- Conviction is weakly preferred to acquittal iff $r - z \geq 0$, i.e. $r \geq z$.
- Acquittal is weakly preferred iff $r \leq z$.

So z acts as a “reasonable doubt” threshold:

$$\text{Juror prefers conviction if } r \geq z, \quad \text{and acquittal if } r \leq z.$$

Diagnostic commentary

LLM Pitfall Alert. A naive model might:

- Swap r and $1 - r$ and derive a nonsensical threshold such as $r \geq 1 - z$.
- Forget that payoffs are *negative* (costs) and compare the wrong quantities.

Student-Level Pitfall. Students frequently mislabel type I and type II errors, or accidentally apply the cost z to acquitting the guilty and $(1 - z)$ to convicting the innocent, reversing the problem's definition.

1.4 Feasible Range of the Correlation Parameter ξ

Main solution

We require all entries in both conditional tables to lie in $[0, 1]$. The sums in each table are already 1 by construction, so we only need non-negativity.

Constraints from $\theta = G$. All four probabilities must be ≥ 0 :

$$\begin{aligned} p^2 + \xi &\geq 0 &\Rightarrow &\xi \geq -p^2, \\ p(1 - p) - \xi &\geq 0 &\Rightarrow &\xi \leq p(1 - p), \\ (1 - p)^2 + \xi &\geq 0 &\Rightarrow &\xi \geq -(1 - p)^2. \end{aligned}$$

Thus, from $\theta = G$,

$$\xi \geq \max\{-p^2, -(1 - p)^2\}, \quad \xi \leq p(1 - p).$$

Since $p \in (1/2, 1)$, we have $p > 1 - p$, so $p^2 > (1 - p)^2$ and hence $-p^2 < -(1 - p)^2$. Therefore

$$\max\{-p^2, -(1 - p)^2\} = -(1 - p)^2,$$

so the constraints from $\theta = G$ reduce to

$$\xi \in [-(1 - p)^2, p(1 - p)].$$

Constraints from $\theta = I$. Similarly, we require:

$$\begin{aligned} (1 - q)^2 + \xi &\geq 0 &\Rightarrow &\xi \geq -(1 - q)^2, \\ (1 - q)q - \xi &\geq 0 &\Rightarrow &\xi \leq (1 - q)q, \\ q^2 + \xi &\geq 0 &\Rightarrow &\xi \geq -q^2. \end{aligned}$$

Thus, from $\theta = I$,

$$\xi \geq \max\{-(1 - q)^2, -q^2\}, \quad \xi \leq (1 - q)q.$$

Since $q \in (1/2, 1)$, we have $q > 1 - q$, so $q^2 > (1 - q)^2$ and $-q^2 < -(1 - q)^2$. Therefore

$$\max\{-(1 - q)^2, -q^2\} = -q^2,$$

and the constraints from $\theta = I$ are

$$\xi \in [-q^2, (1 - q)q].$$

Intersection. The feasible set for ξ must satisfy both sets of constraints, so

$$\xi \in [\underline{\xi}, \bar{\xi}] \equiv [\max\{-(1 - p)^2, -q^2\}, \min\{p(1 - p), (1 - q)q\}].$$

Equivalently,

$$\xi \in \left[-\min\{(1 - p)^2, q^2\}, \min\{p(1 - p), (1 - q)q\} \right].$$

Diagnostic commentary

LLM Pitfall Alert. A common shortcut mistake is to:

- Impose $\xi \in [-1, 1]$ “because it’s a correlation,” ignoring the dependence on (p, q) ;
- Or only check a subset of the four constraints in each table, missing the tightest bound.

Student-Level Pitfall. Students sometimes forget to intersect the constraints from $\theta = G$ and $\theta = I$, thereby allowing values of ξ that make one of the tables invalid.

1.5 Symmetric Monotone Equilibrium with Independent Signals ($\xi = 0$)

Main solution

Set $\xi = 0$, so conditional on θ the signals are i.i.d.

We consider the symmetric monotone strategy profile:

$$\sigma(A) = 1, \quad \sigma(E) = 0,$$

i.e. each juror votes to convict on A and acquit on E .

We must find conditions on z under which this profile is a Bayesian Nash equilibrium (BNE).

Signal structure with $\xi = 0$. Conditional on θ :

- If $\theta = G$:

$$\Pr(A | G) = p, \quad \Pr(E | G) = 1 - p.$$

Hence

$$\Pr(A, A | G) = p^2, \quad \Pr(A, E | G) = \Pr(E, A | G) = p(1 - p), \quad \Pr(E, E | G) = (1 - p)^2.$$

- If $\theta = I$:

$$\Pr(A | I) = 1 - q, \quad \Pr(E | I) = q,$$

so

$$\Pr(A, A | I) = (1 - q)^2, \quad \Pr(A, E | I) = \Pr(E, A | I) = (1 - q)q, \quad \Pr(E, E | I) = q^2.$$

Best response at signal A . Fix the other juror’s strategy as σ , and consider juror i .

At signal $s_i = A$:

- If $s_j = E$, the other juror votes 0, so the verdict is acquittal regardless of i ’s action.
- If $s_j = A$, the other juror votes 1: $(1, 1)$ yields conviction, $(0, 1)$ acquittal. Here i is pivotal.

Thus we need to compare conviction vs acquittal conditional on the event $(s_i, s_j) = (A, A)$. Let

$$r_{AA} := \Pr(\theta = G | s_1 = A, s_2 = A).$$

By the analysis in 1.3, at any information set with posterior r ,

$$\mathbb{E}[u | \text{convict}] - \mathbb{E}[u | \text{acquit}] = r - z.$$

So no profitable deviation at A requires

$$r_{AA} \geq z.$$

Compute r_{AA} by Bayes’ rule:

$$\begin{aligned} \Pr(\theta = G, A, A) &= \mu \Pr(A, A | G) = \mu p^2, \\ \Pr(\theta = I, A, A) &= (1 - \mu) \Pr(A, A | I) = (1 - \mu)(1 - q)^2. \end{aligned}$$

Hence

$$r_{AA} = \frac{\mu p^2}{\mu p^2 + (1 - \mu)(1 - q)^2}.$$

Thus the incentive constraint at A is

$$z \leq \frac{\mu p^2}{\mu p^2 + (1 - \mu)(1 - q)^2}.$$

Best response at signal E . At signal $s_i = E$:

- If $s_j = E$, the other juror votes 0 and the verdict is acquittal regardless of i 's action.
- If $s_j = A$, the other juror votes 1: $(0, 1)$ gives acquittal, $(1, 1)$ conviction, so i is pivotal at (E, A) .

Let

$$r_{EA} := \Pr(\theta = G \mid s_1 = E, s_2 = A).$$

To prevent a deviation from acquit to convict at E , we require that convict be no better than acquit, i.e. $r_{EA} \leq z$.

Compute r_{EA} :

$$\begin{aligned} \Pr(\theta = G, E, A) &= \mu \Pr(E, A \mid G) = \mu p(1 - p), \\ \Pr(\theta = I, E, A) &= (1 - \mu) \Pr(E, A \mid I) = (1 - \mu)q(1 - q). \end{aligned}$$

Thus

$$r_{EA} = \frac{\mu p(1 - p)}{\mu p(1 - p) + (1 - \mu)q(1 - q)}.$$

Equilibrium condition. The symmetric monotone strategy profile is a BNE if and only if

$$r_{EA} \leq z \leq r_{AA}.$$

i.e.

$$\frac{\mu p(1 - p)}{\mu p(1 - p) + (1 - \mu)q(1 - q)} \leq z \leq \frac{\mu p^2}{\mu p^2 + (1 - \mu)(1 - q)^2}.$$

Diagnostic commentary

LLM Pitfall Alert. A naive LLM may:

- Ignore pivotality and use $\Pr(G \mid s_i)$ instead of $\Pr(G \mid s_i, s_j)$.
- Treat the juror as directly choosing the verdict and apply the threshold $r \geq z$ to $\Pr(G \mid A)$ and $\Pr(G \mid E)$, missing the unanimity rule.

Student-Level Pitfall. A common mistake is to check best responses only at the level of individual signals (e.g. “if I see E , I should acquit if $\Pr(G \mid E) \leq z$ ”), without restricting attention to the states where the juror is actually pivotal, i.e. where changing their vote changes the verdict.

1.6 Error Probabilities Under the Symmetric Equilibrium ($\xi = 0$)

Main solution

Under $\xi = 0$ and the symmetric strategy $\sigma(A) = 1, \sigma(E) = 0$, the mapping from signals to actions is:

$$(A, A) \mapsto (1, 1), \quad (A, E) \mapsto (1, 0), \quad (E, A) \mapsto (0, 1), \quad (E, E) \mapsto (0, 0).$$

Joint pmf over (θ, a_1^*, a_2^*) . For $\theta = G$:

$$\begin{aligned}\Pr(\theta = G, a_1^* = 1, a_2^* = 1) &= \mu \Pr(A, A | G) = \mu p^2, \\ \Pr(\theta = G, a_1^* = 1, a_2^* = 0) &= \mu \Pr(A, E | G) = \mu p(1 - p), \\ \Pr(\theta = G, a_1^* = 0, a_2^* = 1) &= \mu \Pr(E, A | G) = \mu p(1 - p), \\ \Pr(\theta = G, a_1^* = 0, a_2^* = 0) &= \mu \Pr(E, E | G) = \mu(1 - p)^2.\end{aligned}$$

For $\theta = I$:

$$\begin{aligned}\Pr(\theta = I, a_1^* = 1, a_2^* = 1) &= (1 - \mu) \Pr(A, A | I) = (1 - \mu)(1 - q)^2, \\ \Pr(\theta = I, a_1^* = 1, a_2^* = 0) &= (1 - \mu) \Pr(A, E | I) = (1 - \mu)(1 - q)q, \\ \Pr(\theta = I, a_1^* = 0, a_2^* = 1) &= (1 - \mu) \Pr(E, A | I) = (1 - \mu)q(1 - q), \\ \Pr(\theta = I, a_1^* = 0, a_2^* = 0) &= (1 - \mu) \Pr(E, E | I) = (1 - \mu)q^2.\end{aligned}$$

Type I and Type II error probabilities. *Type I error* = innocent convicted: $(\theta = I, a_1^* = a_2^* = 1)$. Thus

$$\Pr(\text{Type I error}) = \Pr(\theta = I, a_1^* = 1, a_2^* = 1) = (1 - \mu)(1 - q)^2.$$

Type II error = guilty acquitted: $(\theta = G, (a_1^*, a_2^*) \neq (1, 1))$. So

$$\begin{aligned}\Pr(\text{Type II error}) &= \Pr(\theta = G) \Pr(\text{acquittal} | G) \\ &= \mu[1 - \Pr(A, A | G)] = \mu(1 - p^2).\end{aligned}$$

Equivalently, by summing the three acquittal profiles:

$$\Pr(\text{Type II error}) = \mu[2p(1 - p) + (1 - p)^2] = \mu(1 - p^2).$$

Diagnostic commentary

LLM Pitfall Alert. A naive model may compute $\Pr(a_i = 1, \theta = I)$ as “type I error” or $\Pr(a_i = 0, \theta = G)$ as “type II error,” ignoring that the relevant object is the *committee’s verdict*, not each individual vote.

Student-Level Pitfall. Students often forget that under unanimity, acquittal occurs in three action profiles: $(0, 0)$, $(1, 0)$ and $(0, 1)$, not only $(0, 0)$.

1.7 Correlated Signals: Equilibrium and Error Probabilities

Now consider $\xi \in [\underline{\xi}, \bar{\xi}] \setminus \{0\}$, with the same symmetric strategy $\sigma(A) = 1, \sigma(E) = 0$.

Main solution

Conditional distributions with correlation. For $\theta = G$:

	$s_2 = A$	$s_2 = E$
$s_1 = A$	$p^2 + \xi$	$p(1 - p) - \xi$
$s_1 = E$	$p(1 - p) - \xi$	$(1 - p)^2 + \xi$

For $\theta = I$:

	$s_2 = A$	$s_2 = E$
$s_1 = A$	$(1 - q)^2 + \xi$	$(1 - q)q - \xi$
$s_1 = E$	$(1 - q)q - \xi$	$q^2 + \xi$

The mapping from signals to actions remains:

$$(A, A) \mapsto (1, 1), \quad (A, E) \mapsto (1, 0), \quad (E, A) \mapsto (0, 1), \quad (E, E) \mapsto (0, 0).$$

(i) Equilibrium conditions

As before, we fix the other juror's strategy and check best responses at each signal s_i for juror i .

Best response at A .

Juror i is pivotal when $(s_i, s_j) = (A, A)$. Let

$$r_{AA}(\xi) := \Pr(\theta = G \mid s_1 = A, s_2 = A).$$

No profitable deviation from convict at A requires

$$r_{AA}(\xi) \geq z.$$

Using Bayes' rule:

$$\begin{aligned} \Pr(\theta = G, A, A) &= \mu(p^2 + \xi), \\ \Pr(\theta = I, A, A) &= (1 - \mu)((1 - q)^2 + \xi), \end{aligned}$$

so

$$r_{AA}(\xi) = \frac{\mu(p^2 + \xi)}{\mu(p^2 + \xi) + (1 - \mu)((1 - q)^2 + \xi)}.$$

The incentive constraint at A is

$$z \leq r_{AA}(\xi).$$

Best response at E .

Juror i is pivotal when $(s_i, s_j) = (E, A)$. Let

$$r_{EA}(\xi) := \Pr(\theta = G \mid s_1 = E, s_2 = A).$$

To prevent a deviation from acquit to convict at E , we require $r_{EA}(\xi) \leq z$.

Compute:

$$\begin{aligned} \Pr(\theta = G, E, A) &= \mu(p(1 - p) - \xi), \\ \Pr(\theta = I, E, A) &= (1 - \mu)(q(1 - q) - \xi), \end{aligned}$$

so

$$r_{EA}(\xi) = \frac{\mu(p(1 - p) - \xi)}{\mu(p(1 - p) - \xi) + (1 - \mu)(q(1 - q) - \xi)}.$$

The incentive constraint at E is

$$z \geq r_{EA}(\xi).$$

Equilibrium condition. The symmetric monotone strategy profile remains a BNE if and only if

$$r_{EA}(\xi) \leq z \leq r_{AA}(\xi),$$

with $r_{EA}(\xi)$ and $r_{AA}(\xi)$ as above. When $\xi = 0$, these reduce to the conditions in 1.5.

(ii) Type I and Type II error probabilities

Type I error (innocent convicted). Type I error occurs when $\theta = I$ and both jurors convict, i.e. $(a_1^*, a_2^*) = (1, 1)$, which corresponds to $(s_1, s_2) = (A, A)$. Hence

$$\Pr(\text{Type I error}) = \Pr(\theta = I, A, A) = (1 - \mu)((1 - q)^2 + \xi).$$

Type II error (guilty acquitted). Type II error occurs when $\theta = G$ and the verdict is acquittal, i.e. $(a_1^*, a_2^*) \neq (1, 1)$. Under $\theta = G$, conviction occurs only at (A, A) , whose probability is $p^2 + \xi$, so acquittal has probability $1 - (p^2 + \xi)$. Therefore

$$\Pr(\text{Type II error}) = \Pr(\theta = G, \text{acquittal}) = \mu(1 - p^2 - \xi).$$

(iii) Comparison and the role of ξ

For $\xi = 0$:

$$\Pr(\text{Type I}) = (1 - \mu)(1 - q)^2, \quad \Pr(\text{Type II}) = \mu(1 - p^2).$$

For general ξ :

$$\Pr(\text{Type I}) = (1 - \mu)((1 - q)^2 + \xi), \quad \Pr(\text{Type II}) = \mu(1 - p^2 - \xi).$$

Thus:

- If $\xi > 0$ (positive correlation), (A, A) and (E, E) become more likely, and disagreement (A, E) , (E, A) less likely. Then:

$$\Pr(\text{Type I}) \uparrow, \quad \Pr(\text{Type II}) \downarrow.$$

The committee becomes more prone to wrongful conviction of innocents, but less prone to wrongful acquittal of the guilty.

- If $\xi < 0$ (negative correlation), disagreement signals become more likely, leading to

$$\Pr(\text{Type I}) \downarrow, \quad \Pr(\text{Type II}) \uparrow.$$

So ξ governs a trade-off between type I and type II errors: more correlated evidence increases the probability of unanimous strong signals (and hence conviction), shifting mass from type II to type I errors.

Diagnostic commentary

LLM Pitfall Alert. Typical mistakes:

- Using the independent-signal posteriors from 1.5 unchanged, even when $\xi \neq 0$.
- Computing error probabilities as if only the marginal probabilities changed, without adjusting the (A, A) and (E, E) cells for ξ .

Student-Level Pitfall. It's easy to think that positive correlation “obviously” makes (A, A) stronger evidence of guilt. In reality, $r_{AA}(\xi)$ depends on how (A, A) probabilities change in both states G and I , so the effect on posteriors and on the equilibrium range for z can be subtle, even though the effect on error probabilities is linear and transparent.

Remark (Connection to Feddersen–Pesendorfer). When $\xi = 0$, this committee problem is closely related to the jury models of Feddersen and Pesendorfer: signals are conditionally i.i.d., the unanimity rule makes jurors care about pivotal events (such as (A, A) or (E, A)), and the range of z for which “convict on A , acquit on E ” forms a BNE is determined by posteriors at those pivotal information sets. The correlation parameter ξ generalizes their information structure to correlated private signals and elucidates how shared evidence affects both incentives and the trade-off between type I and type II errors.